

Digits in Disguise — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 • MCQ**[1]**

In a cryptarithm, each letter stands for:

- (a) a single digit (same letter = same digit) (b) any number (c) an operation (d) a place value

Q2 • MCQ**[1]**

If the number 732b is divisible by 5, then b can be:

- (a) 0 only (b) 5 only (c) 0 or 5 (d) any digit

Q3 • FILL IN THE BLANK**[1]**

In the cryptarithm $A B \times A = C C C$ with solution $37 \times 3 = 111$, the digit C is _____.

Q4 • MCQ**[1]**

For 31z5 to be a multiple of 9, the smallest value of z is:

- (a) 0 (b) 3 (c) 6 (d) 9

Q5 • TRUE / FALSE**[1]**

In a cryptarithm, the same letter always represents the same digit.

True False

Q6 · ASSERTION-REASON**[1]**

Assertion (A): The number $6a2$ is divisible by 3 when $8 + a$ is a multiple of 3.

Reason (R): A number is divisible by 3 when the sum of its digits is a multiple of 3.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7**[2]**

Find all digits a so that $6a2$ is divisible by 3.

Q8**[2]**

Solve the cryptarithm $A B \times A = C C C$. Find A, B and C.

Q9**[2]**

Find all digits b so that $732b$ is divisible by 6.

Long answer (3 marks each)

Q10**[3]**

Find all values of z so that $31z5$ is a multiple of 9, and write the two numbers obtained.

Q11

[3]

Find all digits b so that the 6-digit number $4732b0$ is divisible by 4.

Q12

[3]

Solve $A \times BC = DDD$, where A, B, C, D are digits and DDD is a three-digit repdigit. Give one valid solution and check it.

Total: 21 marks · Check your work against the [answer key](#).